

☆☆ Question 我的问题是, 随机变量 X 为所有能在只从性质量度所有事件分析
 下, 又能完备的保证所有事件均能以此种方式量度?

1. 函 R, S 限, 将 ω 中每个切符定义进行量度, 而实数域是主要的.
2. 信息有损是必然的 (将复杂事件 ω 映射为 $\mathbb{R}$), 完备性的真諦在于“足够好”.
 没有模型是绝对完美的, “mcs” — 2015/5/18 — 1:43 — page 739 — #747 只要她非逻辑的精度, 并能
 建立有义的理论, 我是“足够”的.
3. 如果第一级度不够, 自然可以定义更多的随机变量. 其中由所有随机变量组成

18 Random Variables

4. 随机变量并非普通函数, 原 $\Rightarrow$ Thus far, we have focused on probabilities of events. For example, we computed the probability that you win the Monty Hall game or that you have a rare medical condition given that you tested positive. But, in many cases we would like to know more. For example, *how many* contestants must play the Monty Hall game until one of them finally wins? *How long* will this condition last? *How much* will I lose gambling with strange dice all night? To answer such questions, we need to work with random variables.

是可测函数 $\Rightarrow$ 对于轴上每个 ω 我构造出事件, 其 $\Rightarrow$ 每个样本空间中必须 $\Rightarrow$ 是一个全体的事件 (即属于 $\mathcal{F}$ -代数).

$\Rightarrow$ 即从样本空间出发, 有 $\omega \rightarrow R \rightarrow R(\omega)$; 及值域同构, 日子集均有一个唯一的 ω 的半序.

18.1 Random Variable Examples

所以并非所有随机变量是 $\Rightarrow$ Definition 18.1.1. A random variable R on a probability space is a total function whose domain is the sample space. $\Rightarrow$ 随机变量实际上是函数.

是事件, 而是值域中的 $\Rightarrow$ The codomain of R can be anything, but will usually be a subset of the real numbers. Notice that the name “random variable” is a misnomer; random variables are actually functions.

个事件都于样本空间中 $\Rightarrow$ For example, suppose we toss three independent, unbiased coins. Let C be the number of heads that appear. Let $M = 1$ if the three coins come up all heads or all tails, and let $M = 0$ otherwise. Now every outcome of the three coin flips uniquely determines the values of C and M . For example, if we flip heads, tails, heads, then $C = 2$ and $M = 0$. If we flip tails, tails, tails, then $C = 0$ and $M = 1$. In effect, C counts the number of heads, and M indicates whether all the coins match.

明确定义, 即有结构 $\Rightarrow$ Since each outcome uniquely determines C and M , we can regard them as functions mapping outcomes to numbers. For this experiment, the sample space is:

完备, $\Rightarrow$ 引出问题: R 如何产生, $S = \{HHH, HHT, HTH, HTT, THH, THT, TTH, TTT\}$.

全? $\Rightarrow$ R 是一个函数, 函数并非公共 $\Rightarrow$ Now C is a function that maps each outcome in the sample space to a number as follows:

而且一种确定的对应法则

$C(HHH) = 3$	$C(THH) = 2$
$C(HHT) = 2$	$C(THT) = 1$
$C(HTH) = 2$	$C(TTH) = 1$
$C(HTT) = 1$	$C(TTT) = 0$

定义: 一个从 $S \rightarrow \mathbb{R}$ 的函 $\Rightarrow$ 这里的规则可以是一个显式公式, 可以是一个表格, 可以是一个算法或判定过的, 不管用何种方式, 只要这个法则确定、点收义, 且对 S 中每个 ω 都有一个值, 那它就是函数.

$\Rightarrow$ 是一个规则, 隐含中的 ω 不 $\Rightarrow$ 而关于如何保证, 这是意义的一部分, 多措地说, $f: \Omega \rightarrow \mathbb{R}$ 的符号性, 并不要求我们就能

有一个数或有一个 y , $f(\omega) = y$

“写出”或“计算”每个值, 其只要求我们于一个绝对集合: $F \subseteq \Omega \times \mathbb{R}$.

满足: 1. 全性: $\forall x \in \Omega$, 都有 $y \in \mathbb{R}$, 使得 $(x, y) \in F$.

2. 单值性: $(x, y_1) \in F$ 且 $(x, y_2) \in F$, 则 $y_1 = y_2$. F 即为 f 的图 (graph).

证明见网页 P740 #748.